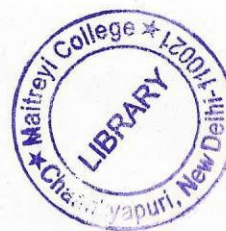


09.06.24(A)

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Unique Paper Code: 32175912

Name of the Paper: GE 7 Molecules of life

Name of the Course: B.Sc. (Hons.)

Semester: II/IV

Duration: 3 hours

Maximum Marks: 75

(Write your Roll No. on the top immediately on receipt of this question paper)

Instructions for Candidates

1. Attempt any *six* questions in all.
2. All questions carry equal marks.

1. (a) What is Ninhydrin reagent? Give the mechanism of α -amino acid with Ninhydrin.
(b) What are disaccharides? Give the structure of lactose and sucrose and mention the linkage
(c) Differentiate between fats and oil. Give the skeletal structure of a lipid which on hydrolysis yields glycerol, stearic acid and comment on its physical properties.
(4,4,4.5)
2. (a) How will you convert?
 - I. Aldohexose into aldopentose
 - II. Glucose into fructose
(b) Write the Merrifield synthesis of the dipeptide Phe-Val. Discuss the advantages of solid phase synthesis.
(c) Write down the structure of glyceryl trioleate. Calculate its iodine number (Molecular weight of glyceryl trioleate : 884, atomic weight of iodine:127). Write its significance.
(4,4,4.5)
3. (a) What is fermentation? Discuss lactic acid and ethanol fermentation.
(b) Explain the effect of temperature and pH on the activity of enzymes.
(c) What is the difference between nucleosides and nucleotides? What is the chemical structure of cytidine and adenosine 5'-monophosphate (AMP)?
(4,4,4.5)
4. (a) Differentiate between reversible and irreversible inhibition of enzyme activity.
(b) What is the structure-activity relationship (SAR) of drug molecules. How does the presence of the -OH and -NH₂ functional groups affect the binding affinity of a drug

molecule to its target receptor?

(c) Explain, how do messenger RNA (mRNA), transfer RNA (tRNA), ribosomal RNA (rRNA) differs in terms of their functions. (4,4,4.5)

5. (a) What are essential and non-essential amino acids with examples?

(b) Draw the structure of a fragment of DNA showing A-T and G-C pairing.

(c) Illustrate the inter-relationship in metabolic pathways of protein, fats and carbohydrates. (4,4,4.5)

6. (a) Draw Fischer and Haworth Projection of β -D-fructofuranose and β -D-glucopyranose.

(b) (i) Write the structure and biological importance of cholesterol.

(ii) Define calorific value of food. What is the standard calorific content of Proteins and Fats?

(c) What do you understand by ω -3 and ω -6 fatty acids. Give name and structure of one fatty acid belonging to each category. (4,4,4.5)

7. (a) Outline the steps involve in Citric acid cycle.

(b) Differentiate between glycolipids and triglycerides. What is their significant role in biological systems?

(c) Define isoelectric point. Calculate isoelectric point of arginine

Given $pK_a(\text{COOH}) = 2.17$, $pK_a(\text{NH}_3^+) = 9.04$, $pK_a(\text{side chain}) = 12.48$

(4,4,4.5)

8. Write short notes on any three of the following: -

(a) Secondary structure of protein

(b) lock and key mechanism

(c) Krebs' Cycle

(d) Amylose and Amylopectin

(4,4,4.5)